From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced), 2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – V
PAPER-1

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	C	C
2.	A	A	B
3.	A	D	C
4.	A	A	C
5.	B	B	B
6.	D	C	D
7.	A	C	A
8.	B	A	A
9.	A, B, C	B, C	B, C
10.	B, C	B, C, D	A, B
11.	C, D	B, D	A, B
12.	B, C, D	A, B, D	A, B, C
13.	A, C, D	A, B	A, B, D
14.	C	C	B
15.	B	B	A
16.	C	B	B
17.	A	C	A
18.	D	C	D
1.	3	9	9
2.	1	7	3
3.	2	6	4
4.	5	7	4
5.	1	8	0
6.	3	5	0
7.	1	6	0
8.	8	3	2
9.	5	5	7
10.	0	4	2

Physics

PART – I

SECTION – A

2. $\frac{9}{4}$ intensity of S_1 = 9 times of intensity S_2

Intensity of maxima = $\left(\sqrt{I_1}+\sqrt{I_2}\right)^2 = 9\,I_2$

$\Rightarrow$ $S_2P - S_1P = n\lambda$ for maxima

$\sqrt{d^2+x^2} - x = n\lambda$ $\qquad x = \frac{d^2-n^2\lambda^2}{2n\lambda}$

3. $\sin\alpha = \frac{\ell}{4y_0^2}$, $\alpha = \sin^{-1}\left(\frac{\ell}{4y_0^2}\right)$

4. $\frac{2mv_0}{qB_0} = \frac{1}{2}\left(\frac{qE_0}{m}\right)t^2$

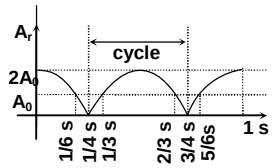
$\frac{4m^2v_0}{q^2E_0B_0} = t^2$

$x = v_0 \times \frac{2m}{q}\sqrt{\frac{v_0}{E_0B_0}}$

6. $y_1 = A\sin\omega_1 t$

$y_2 = A\sin\omega_2 t$

$y_r = 2A\cos\left\{\frac{(\omega_2-\omega_1)}{2}t\right\}\left\{\sin\frac{(\omega_2+\omega_1)}{2}t\right\}$



Resultant amplitude $A_r = 2A_0|\cos(\Delta\omega)t/2|$

$(\Delta\omega)\frac{t}{2} = \frac{\pi}{2} \Rightarrow t = \frac{1}{4}s$

$(\Delta\omega)\frac{t}{2} = \frac{\pi}{3} \Rightarrow t = \frac{1}{6}s$

In one cycle of intensity of 1/2s, the detector remain idle for

$2\left(\frac{1}{4}-\frac{1}{6}\right)s = \frac{1}{6}\text{sec}$

$\therefore$ In ½ sec cycle, active time is $\left(\frac{1}{2}-\frac{1}{6}\right)$ = 1/3 sec

$\therefore$ In 12 sec interval, active time is $12\times\frac{(1/3)}{(1/2)}$ = 8 sec

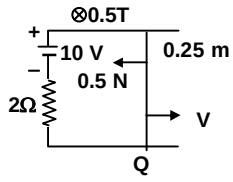
7. Let ϕ_1 = 4eV, then ϕ_2 = 2eV

$(E - \phi)$ represent kinetic energy of most energetic electron.

$E - \phi_2 = 2(E - \phi_1)$

$\Rightarrow E = 6eV$

8. Induced e.m.f. = $B\ell v = 0.125$ V

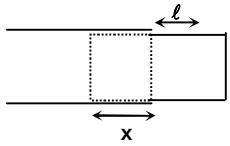


current $I = \frac{10 - e}{R} = \frac{10 - 0.125\ V}{2}$

Force $BI\ell = 0.5\left(\frac{10 - 0.125}{2}\right)0.25 = 0.5N$ (given)

Solving v = 16 m/s.

11. $C = \frac{\varepsilon_0(a - x)a}{d} + \frac{\varepsilon_0 xak}{d}$



$= \frac{\varepsilon_0 a[a - x + kx]}{d}$

$= \frac{\varepsilon_0 a}{d}[a + (k-1)x]$

$U = \frac{\varepsilon_0 aV^2}{2d}[a + (k-1)x]$

$F = \frac{dU}{dx} = \frac{\varepsilon_0 aV^2(k-1)}{2d}$

$v^2 = v_0^2 + 2Ax = v_0^2 + \frac{\varepsilon_0 aV^2(k-1)}{md}x$

$E = \frac{1}{2}mv_0^2 + \frac{\varepsilon_0 aV^2(k-1)x}{2d} + \frac{\varepsilon_0 aV^2}{2d}[a + (k-1)x]$

$E_{max} = \frac{1}{2}mv_0^2 + \frac{\varepsilon_0 a^2V^2(2k-1)}{2d}$

12. P.D. same & electric field same
Charge will be kC ε
Energy will be ½ kC ε^2

14. $P_0 + 0.8\ \rho_w gh + \rho_w g(2h) + 1.2\ \rho_w g(3h) = P_0 + 1.2\ \rho_w gh_2$

$\Rightarrow 1.2\ h_2 = h[0.8 + 2 + 3.6] = 6.4\ h \Rightarrow h_2 = \frac{6.4}{1.2}h$

$P_0 + 0.8\ \rho_w gh + \rho_w g(2h) = \rho_w gh_1 + P_0$

$\Rightarrow\ h_1 = h[0.8 + 2] \Rightarrow h_1 = 2.8\ h$

$h_2/h_1 = 40/21$

15. $P_0 + \frac{1}{2}\rho_w v^2 = P_0 + 0.8\rho_w gh + \rho_w g(2h)$

$\Rightarrow \frac{v^2}{2} = gh[0.8 + 2]$

$v = \sqrt{2 \times 9.8 \times 0.28 \times 2.8} = 3.92$ m/s

16-18. In the time interval t = 0 sec to t = t_0 sec, till the time. The relative velocity is not zero the nature of friction will be kinetic.

$\vec{u}_{AB} = 0\hat{i} - (20\hat{i}) = 20\ \text{m/sec}(\hat{i})$

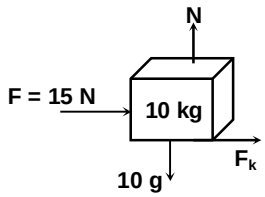
$N = 100\ N \Rightarrow F_k = 0.25 \times 100 = 25\ N$

$\vec{a}_A = 4\text{m/sec}^2(\hat{i})$, $\vec{a}_B = 0\text{m/sec}^2(\hat{i})$

$\vec{a}_{AB} = \vec{a}_A - \vec{a}_B = 4\text{m/sec}^2(\hat{i})$

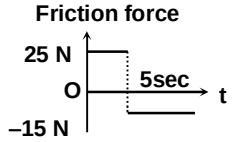
$\vec{V}_{AB} = \vec{u}_{AB} + \vec{a}_{AB}t = \vec{0}$

$\Rightarrow t = 5$ sec



When the relative motion between block A and belt conveyor will be zero, the nature of friction will be static and its magnitude will equal to magnitude of unbalanced external force acting on the block A and its direction will be I the opposite direction of unbalanced external force.

$\Rightarrow F_r = \text{Frictional force} = \begin{cases} 25N(\hat{i}) \Rightarrow \text{kinetic in nature} \\ 15N(-\hat{i}) \Rightarrow \text{Static in nature} \end{cases}$



SECTION –C

1. $\hat{p} = \hat{j}$, $\hat{n} = a\hat{i} - b\hat{j} = \frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}$

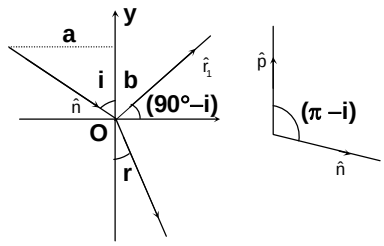
$\hat{r}_1 = \hat{n} - 2(\hat{n}\cdot\hat{p})\hat{p}$

$= (a\hat{i} - b\hat{j}) - 2(-b)\hat{j} = a\hat{i} + b\hat{j}$

$\hat{n}\cdot\hat{p} = \cos(\pi - i) = -b$

$\cos i = b = \frac{\sqrt{3}}{2} \Rightarrow i = 30°$

$\mu = \cot i = \sqrt{3}$



4. For collision the relative velocity of A with respect to B should be along the line AB

6. Using condition of equilibrium

$KL = (m_1 + m_2)g$

Using COME

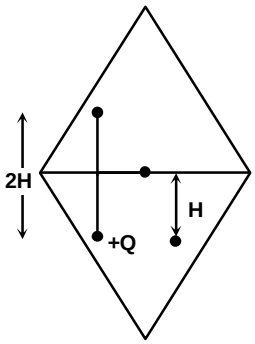
$\frac{1}{2}KL^2 = m_1g(2L)$

$\Rightarrow m_1 = \frac{KL}{4g} = \frac{(m_1 + m_2)}{4}$

$\Rightarrow 3m_1 = m_2 = m_2/m_1 = 3$

8. If we place another similar cone on this one, net flux = $\frac{Q}{\varepsilon_0}$

$\Rightarrow$ Flux with the upper cone = $\frac{Q}{\varepsilon_0} - \frac{3Q}{5\varepsilon_0} = \frac{2Q}{5\varepsilon_0} = \frac{8Q}{20\varepsilon_0}$. This must also be the flux associated with the lower cone when the charge is raised through a height 2H because of symmetry.



9. Velocity of first block before collision,

$v_1^2 = 1^2 - 2(2) \times 0.16$

= 1 - 0.64

v_1 = 0.6 m/s

By conservation of momentum, $2 \times 0.6 = 2\ v_1' + 4\, v_2'$

also $v_2' - v_1' = v_1$ for elastic collision

It gives $v_2' = 0.4$ m/s

$v_1' = -0.2$ m/s

Now distance moved after collision

$s_1 = \frac{(0.4)^2}{2 \times 2}$ & $s_2 = \frac{(0.2)^2}{2 \times 2}$

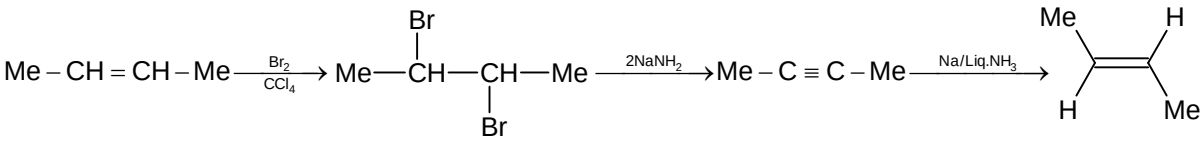
$\therefore\ s = s_1 + s_2 = 0.05$ m = 5 cm.

10. By symmetry, no current will flow in arm along x-axis.

Chemistry

PART – II

3. More activated ring undergo EAS.

4. 

5. 1st titration

Meq. of NaOH $\frac{1}{2}$ Meq. of Na_2CO_3 = Meq. of HCl = $50 \times \frac{1}{10} = 5$

2nd titration

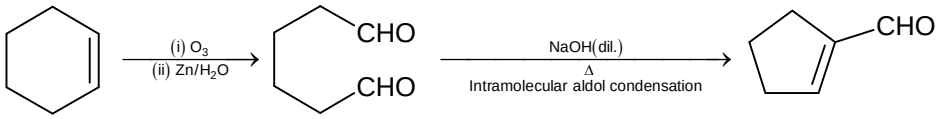
$\frac{1}{2}$ Meq. of Na_2CO_3 = Meq. of HCl = $10 \times \frac{1}{10} = 1$

$\therefore$ Meq. of NaOH = 5 – 1 = 4

$$\text{Wt. of NaOH} = \frac{\text{Meq. of NaOH} \times \text{Equivalent weight}}{1000}$$

$$= \frac{4 \times 40}{1000}$$

= 0.16 g

6. 

7. $8NH_3\,(\text{excess}) + 3Cl_2 \longrightarrow 6NH_4Cl + N_2$

$6NaOH(\text{hot and conc.}) + 3Cl_2 \longrightarrow 5NaCl + NaClO_3 + 3H_2O$

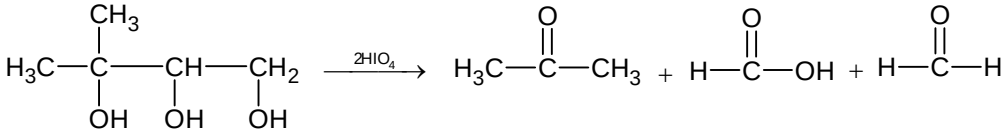
$Fe + 2HCl \longrightarrow FeCl_2 + H_2$

Liberation of H_2 prevents the formation of $FeCl_3$.

Bond dissociation of F – F is less than that of Cl – Cl.

8. $(CH_3)_3SiCl$ is used to stop the growth of polymerisation of linear polymer formed from the hydrolysis of $(CH_3)_2SiCl_2$.

9. Configuration of (B) is (2R, 3S)
Configuration of (C) is (2R, 3S)

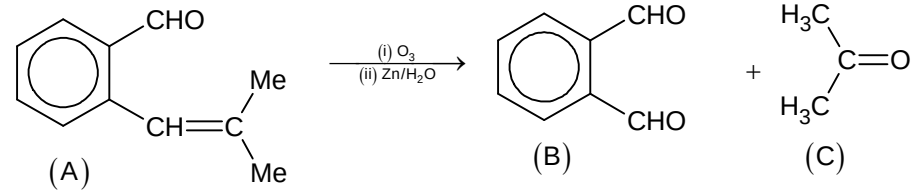
10. 

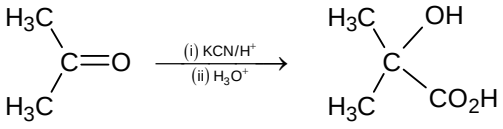
11. 1. Addition of inert gas at constant pressure shifts the equilibrium in that direction which has more number of gaseous moles.
2. Increasing the volume of the container shifts the equilibrium in that direction which has more number of gaseous moles.

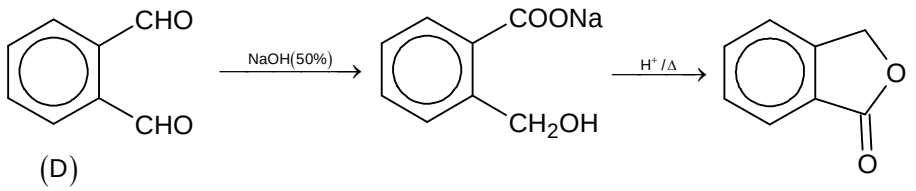
12. Chemisorption unimolecular layer.

13. $[Cu(NH_3)_4]^{2+} \rightarrow dsp^2$
$[Ni(CN)_4]^{2-} \rightarrow dsp^2$
$[CuCl_4]^{2-} \rightarrow sp^3$
$[NiCl_4]^{2-} \rightarrow sp^3$

14.–15.







16.–18.

$$\underset{\text{(White)}}{\overset{(A)}{ZnS}} + H_2SO_4 \longrightarrow \underset{(C)}{ZnSO_4} + \underset{(B)}{H_2S}$$

$$3H_2S + K_2Cr_2O_7 + 4H_2SO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + \underset{(D)}{3S} + 7H_2O$$

$$\underset{(D)}{S} + O_2 \longrightarrow \underset{(E)}{SO_2}$$

$$2H_2S + SO_2 \longrightarrow 2H_2O + 3S$$

$$ZnSO_4 + 2NaOH \longrightarrow Zn(OH)_2 + Na_2SO_4$$

$$Zn(OH)_2 + 2NaOH \longrightarrow \underset{\text{(Soluble)}}{Na_2ZnO_2} + 2H_2O$$

SECTION – C

1. A present in corners and face centre.
B present in alternate tetrahedral voids.

No. of A per unit cell = $6 \times \frac{1}{8} + 6 \times \frac{1}{2} = \frac{15}{4}$

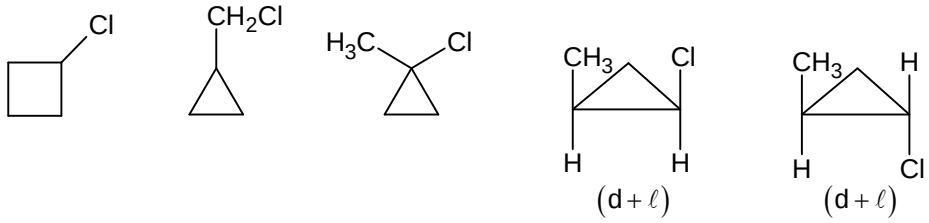
No. of B per unit cell = 4 – 1 = 3

$x : y = \frac{15}{4} : 3 = 5 : 4$

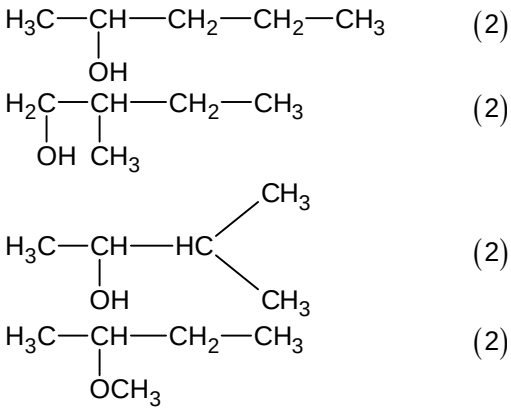
x + y = 5 + 4 = 9

2. All the compounds undergo hydrolysis.

3. Let the number of α-particles emitted be x and the number of β-particles emitted be y.
4x = 238 – 206 = 32
x = 8
2x – y = 10
2 × 8 – y = 10
y = 6.

4. 

Total number of isomers = 7.

5. 

6. $PCl_3 + 3H_2O \longrightarrow \underset{(\text{dibasic})}{H_3PO_3} + 3HCl$

Number of moles required to neutralise H_3PO_3 = 2
Number of moles required to neutralise HCl = 3
Total number of moles of NaOH required to neutralise the solution = 5.

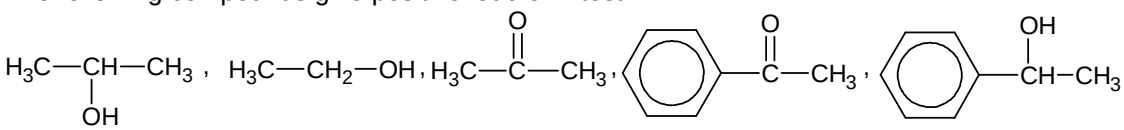
7. $C_xH_{12}(g) + \left(x + \frac{12}{4}\right)O_2(g) \longrightarrow xCO_2(g) + 6H_2O(g)$

Initial moles of gaseous reactant taken = 1 + (x + 3) = x + 4
Final moles of product formed = x + 6
Number of moles increased = 2
Pressure $\propto$ Number of moles.

% of pressure increased $= \frac{2}{(x+4)} \times 100$, $20 = \frac{2}{(x+4)} \times 100$, x = 6.

8. N_2O, NO and CO are neutral oxides remaining are acidic oxides.

9. The following compounds give positive iodoform test



10. In the following compound sulphur has +6 oxidation state
$H_2S_2O_7$, H_2SO_4, H_2SO_5, $H_2S_2O_8$

Mathematics PART – III

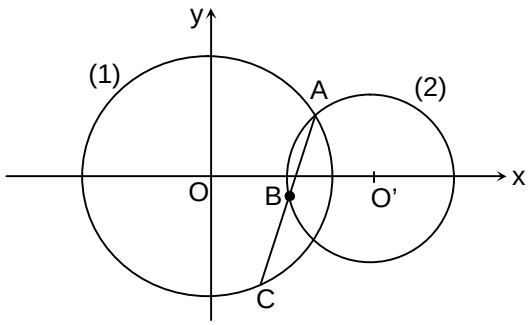
SECTION – A

2. Let origin be centre of larger circle and O' be the centre of smaller circle.

Equation of (1): $x^2 + y^2 = 2$

Equation of (2): $(x-2)^2 + y^2 = 1$

$\therefore A\left(\frac{5}{4}, \frac{\sqrt{7}}{4}\right)$



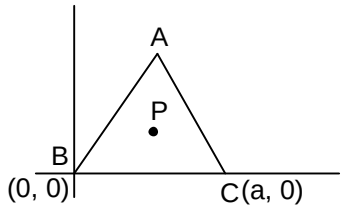
$x_B = \frac{x_A + x_C}{2}, \; y_B = \frac{y_A + y_C}{2}$

$\therefore x_C^2 + y_C^2 = 2$

$\therefore \left(\frac{x_C + \frac{5}{4}}{2} - 2\right)^2 + \left(\frac{y_C + \frac{\sqrt{7}}{4}}{2}\right)^2 = 1$

$\therefore AC = \sqrt{\frac{7}{2}}$

3. Let B (0, 0), C (a, 0), A (h, k) and $P(\alpha, \beta)$.



$\frac{d(A, BC)}{d(P, BC)} = \frac{k}{\beta}$

Equation of AC is $kx - (h-a)y - ka = 0$

$\therefore \frac{d(B, AC)}{d(P, AC)} = \frac{ka}{k\alpha - (h-a)\beta - ka}$

Equation of AB is $kx - hy = 0$

$\frac{d(C, AB)}{d(P, AB)} = \frac{ka}{k\alpha - h\beta}$

Solving, $\alpha = \frac{h+a}{3}, \; \beta = \frac{k}{3}$

4. Ortho–centres of triangles formed by three tangents and corresponding normals to a parabola are equidistant from axis of parabola.

5. Number beginning with 2 and not divisible by 5 is 6! – 120 = 600
Number beginning with 3 and not divisible by 5 = 600
Number beginning with 5 and not divisible by 5 = 720
Total = 1920
Number beginning with 6 2 and not divisible by 5 is 120 – 24 = 96
$\therefore$ Total = 2016
$\therefore$ 2017th number is 6325789

6. $(PQ)^T = Q^T P^T = QP = PQ$

PQ is symmetric

$\therefore PQP^{-1} = QPP^{-1} = Q$

$\Rightarrow P^{-1}PQP^{-1} = P^{-1}Q$

$\Rightarrow QP^{-1} = P^{-1}Q$

$\left(P^{-1}Q\right)^T = Q^T\left(P^{-1}\right)^T = P^{-1}Q$

8. $P(u_i) \propto i^2 + 2$

$\Rightarrow P(u_i) = k\left(i^2 + 2\right)$

$u_1, u_2, \ldots\ldots, u_n$ are mutually exclusive events.

$\therefore P(u_1) + P(u_2) + \ldots\ldots + P(u_n) = 1$

$\Rightarrow k\left[1^2 + 2 + 2^2 + 2 + 3^2 + 2 + \ldots\ldots n^2 + 2\right] = 1$

$\Rightarrow k = \dfrac{6}{n(n+1)(2n+1) + 12n}$

$A \rightarrow$ event of drawing a white ball

$P(A) = k\left(1^2 + 2\right) \times \dfrac{2}{2n+1} + k\left(2^2 + 2\right).\dfrac{4}{2n+1} + k\left(3^2 + 2\right).\dfrac{6}{2n+1} + \ldots\ldots + k\left(n^2 + 2\right).\dfrac{2n}{2n+1}$

$= \dfrac{k}{2n+1}\left[1^2.2 + 2^2.4 + 3^2.6 + \ldots\ldots + n^2.2n + 2.2 + 4.2 + 6.2 + \ldots\ldots + 2n.2\right]$

$= \dfrac{2.k}{2n+1}\left[1^3 + 2^3 + 3^3 + \ldots\ldots + n^3 + 2\left(1 + 2 + 3 + \ldots\ldots + n\right)\right]$

$= \dfrac{2k}{2n+1}\left[\dfrac{n^2(n+1)^2}{4} + n(n+1)\right]$

$= \dfrac{2.6\left[n^2(n+1)^2 + 4n(n+1)\right]}{4(2n+1)\left[n(n+1)(2n+1) + 12n\right]}$

$\underset{n\to\infty}{Lt}\, P(A) = \underset{n\to\infty}{Lt} \dfrac{3\left(1+\dfrac{1}{n}\right)^2 + 4\left(\dfrac{1}{n^2} + \dfrac{1}{n^3}\right)}{\left(2+\dfrac{1}{n}\right)^2\left(1+\dfrac{1}{n}\right) + 12\left(\dfrac{2}{n^2} + \dfrac{1}{n^3}\right)} = \dfrac{3}{4}$

9. $\dfrac{\sin\dfrac{\pi}{12}}{\sin x} + \dfrac{\cos\dfrac{\pi}{12}}{\cos x} = 2$

10. $|z_1| = |z_2| = 1;\ |z_1 + z_2| = 1$

$\Rightarrow \left(z_1 + z_2\right)\left(\bar{z}_1 + \bar{z}_2\right) = 1$

$|z_1 + z_2| = |z_1|\ |1 + e^{i\theta}| = 1$

$|z_2 - z_1| = |z_1|\ |e^{i\theta} - 1|$

11. $\dfrac{100}{2}\left(x_1 + x_{100}\right) = -1$

$\dfrac{50}{2}\left(x_1 + d + x_{100}\right) = 1$

12. Let $y < x$

$\Rightarrow \frac{y}{x} = \frac{x}{x+y}$

$\Rightarrow R = \frac{1}{1+R}$

13. $(a + b) = x$, then $x^3 - 21x + 20 = 0$
Which has roots 1, 4, –5

SECTION – C

8. $2(1+3+5+7+.....+a)+2(1+3+5+7+.....+b)=2(1+3+5+7+.....+c)$

$\Rightarrow \left[2+\left(\frac{a+1}{2}-1\right)2\right]\frac{a+1}{2}+\left[2+\left(\frac{b+1}{2}-1\right)2\right]\frac{b+1}{2}=\left[2+\left(\frac{c+1}{2}-1\right)2\right]\frac{c+1}{2}$

$\Rightarrow (a+1)^2+(b+1)^2=(c+1)^2$

$\because a > 6$ and $a+b+c=21$

$\therefore a = 7, b = 5, c = 9$

$\therefore a-b=2$

9. $p=\frac{{}^3C_1\times {}^8P_6+{}^3C_1\times {}^8P_6-{}^3P_2\times {}^7P_5}{{}^9P_7}=\frac{7}{12}$

10. $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}=\begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

$\Rightarrow a_{11}+2a_{12}=a_{11}+a_{21}$ ..... (i)

$a_{11}+a_{12}=2a_{12}+a_{22}$ ..... (ii)

$a_{21}+2a_{22}=2a_{11}+a_{21}$

$a_{21}+a_{22}=2a_{12}+a_{22}$

$\therefore$ From (ii), $a_{11}=a_{22}$

$a_{21}=2a_{12}$

$\because a_{11}a_{22}=a_{21}a_{21}\Rightarrow (a_{11})^2=2(a_{12})^2$

$\Rightarrow a_{11}=\sqrt{2}a_{12}$